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10.6

14. Fabricating Systems A production plant has two fabricating systems, both of which are maintained at 2-week intervals. However, one system is twice as old as the other. The number of finished products fabricated daily by each of the systems is recorded for 30 working days, with the results given in the table. Do these data present sufficient evidence to conclude that the variability in daily production warrants increased maintenance of the older fabricating system? Use the p-value approach.

New System	Old System
$\bar{x}_1 = 246$	$\bar{x}_2 = 240$
$s_1 = 15.6$	$s_2 = 28.2$

$n_1 = 30$, $s_1 = 15.6$
$n_2 = 30$, $s_2 = 28.2$

① $H_0 : \sigma_o^2 = \sigma_N^2$
$H_a : \sigma_o^2 > \sigma_N^2$

② assume that $\alpha = 0.05$

③ $F = \frac{s_o^2}{s_N^2} = \frac{28.2^2}{15.6^2} = \frac{796.24}{243.36} = 3.268$

$df = 30-1 = 29$

④ $p = P(F_{29,29} \geq 3.268)$
at $0.05 = 1.85$
at $0.005 = 2.66$
$3.268 > 2.66$

⑤ $p < 0.005 < 0.05$ (Reject H_0)

There is sufficient evidence that the older system is more variable, so increased maintenance is justified

16. SAT Scores The SAT subject tests in chemistry and physics for two groups of 15 students each electing to take these tests are given here.

Chemistry	Physics
$\bar{x} = 784$	$\bar{x} = 758$
$s = 114$	$s = 103$
$n = 15$	$n = 15$

To use the two-sample t-test with a pooled estimate of σ^2, you must assume that the two population variances are equal. Test this assumption using the F-test for equality of variances. What is the approximate p-value for the test?

① $H_0 : \sigma_1^2 = \sigma_2^2 \rightarrow \frac{s_1^2}{s_2^2} = 1$
$H_a : \sigma_1^2 \neq \sigma_2^2 \rightarrow \frac{s_1^2}{s_2^2} \neq 1$

② assume, $\alpha = 0.05$

③ $F = \frac{s_1^2}{s_2^2} \sim F_{(n_1-1),(n_2-1)}$

$F = \frac{114^2}{103^2} = \frac{12996}{10609} = 1.2249 \approx 1.225$

$df = 14-1 = 15$

④ $F_{14,14,0.025} = 2.996$

⑤ RR = $F > 2.996$ or $F < -2.996$
$1.225 < 2.996$ (Failed to reject H_0)

p value approx.
at $0.25 = 1.65$
at $0.1 = 2.12$
at $0.05 = 2.48$
$1.225 < 1.65$
p value > 0.25 (one-tail)
p value $> 2 \times 0.25$
p value > 0.5
~approx $p \approx 0.7 > 0.05$ (Fail to reject)

There is no sufficient evidence that the population variances differ, so the assumption of equal variances is reasonable.

22. Stock Risks The closing prices of two common stocks were recorded for a period of 15 days. The means and variances are

$df_1 = df_2 = 14$

$\bar{x}_1 = 40.33$	$\bar{x}_2 = 42.54$
$s_1^2 = 1.54$	$s_2^2 = 2.96$

a. Do these data present sufficient evidence to indicate a difference between the variabilities of the closing prices of the two stocks for the populations associated with the two samples? Give the p-value for the test and interpret its value.

b. Construct a 99% confidence interval for the ratio of the two population variances.

a) ① $H_0 : \sigma_1^2 = \sigma_2^2$
$H_a : \sigma_1^2 \neq \sigma_2^2$

② $F = \frac{s_1^2}{s_2^2} \sim F_{(n_1-1),(n_2-1)}$

③ $F = \frac{2.96}{1.54} = 1.922$

④ p-value approx.
at $0.1 = 2.12$
at $0.25 = 1.65$
$1.922 > 1.65$
$1.65 < 1.922 < 2.12$
$0.25 < \text{p-value} < 0.1$
p-value $= 0.117$ (one tail)
p-value $= 2 \times 0.117 \approx 0.234$

There is no sufficient evidence to conclude that there is a difference between two population variances.

b) 99% CI $\rightarrow \alpha = 0.01/2 = 0.005$

$\left(\frac{s_1^2}{s_2^2}\right)\frac{1}{F_{14,14,0.005}} < \frac{\sigma_1^2}{\sigma_2^2} < \left(\frac{s_1^2}{s_2^2}\right) F_{14,14,0.005}$

$\frac{1.54}{2.96} \cdot \frac{1}{4.25} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{1.54}{2.96} \cdot 4.25$

$0.122 < \frac{\sigma_1^2}{\sigma_2^2} < 2.211$

1 is in the interval, so failed to reject H_0